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# Distribution and conservation of *Acomys cilicicus* (Mammalia: Rodentia) in Turkey

**Ortaç ÇETİNTAŞ[1], Ferhat MATUR[2], Mustafa SÖZEN[1,*]**
[1]Department of Biology, Faculty of Arts and Sciences, Bülent Ecevit University, Zonguldak, Turkey
[2]Department of Biology, Faculty of Arts and Sciences, Dokuz Eylül University, İzmir, Turkey

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**Abstract:** *Acomys cilicicus* is endemic to Turkey and known from a very restricted area. The exact distribution of the species was not known up to now and the IUCN status of the species was Data Deficient (DD). To determine the exact distribution area of the species, 39 localities within the historical distribution were surveyed by using 3243 Sherman traps between 2013 and 2016. Turkish spiny mouse samples were obtained from 14 of these 39 localities and the current distribution of the species was determined. We found that the Turkish spiny mouse has two isolated populations in the area between Silifke and Erdemli with a total distribution area of about 104.5 $km^2$, extending from sea level up to 510 m a.s.l. Population trend estimates showed a steep decline in the last 20 years from 21.42 to 2.75 as trap night index value. These data, along with the decline in habitat quality and continuing threats to the species, merit an IUCN status of Critically Endangered (CR). The main threats for this species are habitat loss due to urbanization, new motorway construction, stone quarry development, conversion of Mediterranean shrublands into agricultural fields, and afforestation. For conservation purposes, a species protection action plan is necessary immediately.

**Key words:** IUCN status, Turkish spiny mouse, habitat changes, threats

## 1. Introduction

The genus *Acomys* is distributed around the Arabian Peninsula, Africa, southern Turkey, Cyprus, and Crete (Bates, 1994). The genus *Acomys* is represented by 19 species (*A. cahirinus*, *A. cilicicus*, *A. cinneraceus*, *A. ignitus*, *A. kempi*, *A. louisae*, *A. minous*, *A. mullah*, *A. nesiotes*, *A. percivali*, *A. russatus*, *A. spinossissimus*, *A. subspinasus*, *A. wilsoni*, *A. airensis*, *A. chudeaui*, *A. dimidiatus*, *A. johannis*, and *A. seurati*) (Wilson and Reeder, 2005).

The taxonomy of the species has been widely discussed and no definite decision has been reached yet. The spiny mouse from Silifke was first reported as *Acomys cahirinus* (Lehmann, 1966; Corbet, 1978), and Lehmann (1969) stated that it might be a part of *A. nesiotes*, and then it was reported as *A. dimidiatus* (Atallah, 1978; Morlok, 1978). Later Kummerloeve (1975) reported it as a member of the *cahirinus-dimidiatus* group. However, Spitzenberger (1978) compared morphological peculiarities of specimens from Silifke with those of *A. nesiotes* from Cyprus and *A. minous* from Crete and classified the Silifke specimens as *Acomys cilicicus* based on the differences in incisors, molar characteristics, hind foot, and body measurements. Though this recognition, Doğramacı (1989) and Harrison and Bates (1991) reported the species in Turkey as *A. cahirinus*. However, in recent years, most authors accepted the name *cilicicus* (Janecek et al., 1991; Bates, 1994; Denys et al., 1994; Macholán et al., 1995; Kurtonur et al., 1996; Kıvanç et al., 1997, 2013; Barome et al., 2000, 2001; Kryštufek and Vohralík, 2001; Musser and Carleton, 2005; Yiğit et al., 2006; Arslan et al., 2008). On the other hand, recently Kryštufek and Vohralík (2009) used the name *Acomys cahirinus cilicicus*. One interesting point is that to explain the distribution pattern of *Acomys cahirinus* and *A. cilicicus* by traditional zoogeographic history is not easy, since no historical connections are available among these localities during *Acomys* dispersion times. Thus, the evolution and dispersion of *Acomys* in Turkey, Cyprus, and Crete was evaluated by Barome et al. (2000) as man-aided dispersion of *A. cahirinus*. In Turkey the spiny mouse population is restricted to a very small area, and genetic drift and adaptation to this habitat seem to have produced the distinguished traits that are used for species identification of *Acomys cilicicus*.

All specimens evaluated from Turkey were collected from Kumkuyu (Lehmann, 1966); 17 km east of Silifke (Spitzenberger, 1978); 20 km east of Silifke (Lehmann, 1969; Kıvanç et al., 1997, 2013); ca. 30 km northeast of Silifke (Morlok, 1978); Narlıkuyu Canyon; and Kızkalesi, Mersin (Macholán et al., 1995). Thus, all known distribution records from Turkey were restricted to the

* Correspondence: spalaxtr@hotmail.com

area between 5 km east and 30 km northeast of Silifke. All of these records were given from sea level to about 100 m above sea level. To date, no study attempted to clarify the total distribution area and population density of this species in Turkey. Moreover, this species has been listed as Data Deficient by the IUCN Red List, mostly because of taxonomic issues, as the IUCN's webpage indicates that this taxon is likely to be conspecific with the widespread species *Acomys cahirinus* and indicates that more research is needed for confirmation. However, most researchers have indicated that this is a different species, as mentioned above.

For proper assessment and conservation actions, establishing the distribution area of a species is a very crucial step. That is why, in the present study, we aimed to determine the distribution area of the endemic Turkish spiny mouse, *Acomys cilicicus*, as well as to contribute to its IUCN evaluation by supplying more data on threat factors, protection priorities, and population trend changes.

## 2. Materials and methods

To determine the distribution area of the Turkish spiny mouse, *Acomys cilicicus*, known distribution points given in the literature and surrounding areas were trapped by Sherman type traps (folding type trap: 7.62 × 8.89 × 22.86 cm) between April 2013 and May 2016. Traps were baited with commercial chocolate balls, peanuts, and corn. A total of 39 sampling points were set (Figure 1) with a

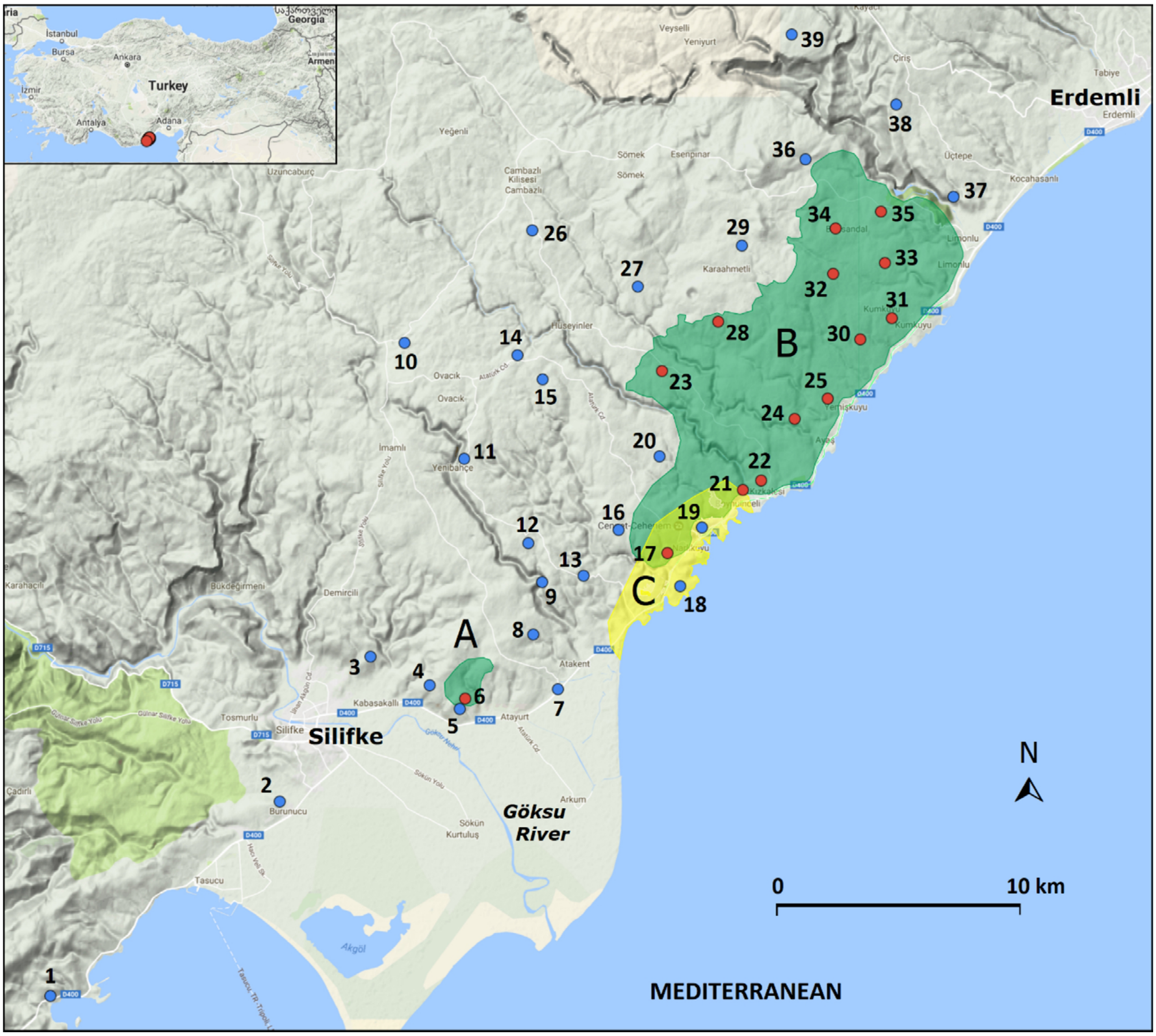


**Figure 1.** Sampling localities. The numbers of localities are the same as in Table 1. Blue circles indicate trapping localities where no *Acomys cilicicus* samples were collected and red circles indicate the localities where *Acomys cilicicus* samples were collected. Green dashed areas show the estimated distribution areas of two isolated populations (A and B on map) of *Acomys* in the study area, and yellow dashed area (C on map) shows the IUCN distribution map of *Acomys cilicicus*.

total of 3243 traps. Between 70 and 100 traps were set at the sampling points on each sampling day. Two of the localities were to the west of the Göksu River whereas the other 37 were to the east of the Göksu River (Figure 1). The trap night index (TNI, number of records/100 trap nights) was calculated according to Gurnell and Flowerdew (2006) for all species captured during the study. The relative abundance map of species collected was computed using the Tableau Version 9.2 program. Each trap was recorded by hand-held GPS and records were transferred to Google Maps to obtain the distribution map (Figure 1). Major threats determined during field surveys were recorded and were classified according to the IUCN Threats Classification Scheme (Version 3.2) (IUCN, 2017). To determine habitat alterations throughout time, we compared Google Earth from 2004 and 2017. All data obtained during the study were evaluated by IUCN threat criteria (IUCN Standards and Petitions Subcommittee, 2016) to determine the IUCN threat status of the species.

## 3. Results and discussion

### 3.1. Distribution

From the 39 localities surveyed (Figure 1; Table 1), we obtained individuals of *Acomys cilicicus* in 14 of them (red circles in Figure 1). These sampling localities show that the current distribution of the Turkish spiny mouse is an area between 5 km E and 32 km NE of Silifke from sea level up to 510 m a.s.l. The distribution reaches about 8 km inland towards the northeastern side. Moreover, the distribution of the species is fragmented, as a small isolated population is present in the north of Esenbel village (number 1 in Figure 1). Traps surrounding this locality did not supply any additional samples. The distribution of the species between Narlıkuyu and Limonlu extends continuously for about 18–19 km (number 2 in Figure 1). According to distribution records supplied here, the total distribution area of the Turkish spiny mouse is about 104.5 km$^2$ around the Silifke region. The estimated distribution area of the isolated population (population A, Figure 1) is about

**Table 1.** Coordinates and altitudes of the localities surveyed and samples collected (+) for *Acomys cilicicus*.

| Localities | Coordinates | Altitude (m a.s.l.) | Sample: yes (+), no (-) |
|---|---|---|---|
| 1. 15 km W of Silifke | 36°16′37.29″N, 33°48′50.61″E | 85 m | - |
| 2. Burunucu | 36°20′58.94″N, 33°55′13.48″E | 60 m | - |
| 3. 3 km E of Silifke | 36°24′13.70″N, 33°57′45.08″E | 208 m | - |
| 4. Kabasakallı | 36°23′35.29″N, 33°59′24.44″E | 200 m | - |
| 5. Esenbel1 | 36°23′3.66″N, 34°0′14.54″E | 105 m | - |
| 6. Esenbel2 | 36°23′18.93″N, 34°0′23.32″E | 148 m | + |
| Esenbel2 | 36°23′18.84″N, 34°0′22.47″E | 149 m | + |
| 7. Karadedeli | 36°23′30.06″N, 34°2′58.42″E | 2 m | - |
| 8. 3 km N of Atakent | 36°24′43.91″N, 34°2′17.36″E | 200 m | - |
| 9. 3 km N of Susanoğlu | 36°25′53.74″N, 34°2′32.15″E | 169 m | - |
| 10. Keşlitürkmenli | 36°31′15.66″N, 33°58′42.23″E | 870 m | - |
| 11. Yenibahçe | 36°28′40.07″N, 34°0′21.86″E | 640 m | - |
| 12. Türkmenuşağı | 36°26′46.56″N, 34°2′9.30″E | 390 m | - |
| 13. 3 km N of Susanoğlu | 36°26′3.00″N, 34°3′41.52″E | 228 m | - |
| 14. Hüseyinler3 | 36°30′26.19″N, 34°2′33.26″E | 650 m | - |
| 15. Hüseyinler1 | 36°30′59.02″N, 34°1′50.75″E | 720 m | - |
| 16. 3 km N of Narlıkuyu | 36°27′4.23″N, 34°4′39.66″E | 291 m | - |
| 17. Narlıkuyu1 | 36°26′27.42″N, 34°5′58.71″E | 88 m | + |
| Narlıkuyu1 | 36°26′28.80″N, 34°5′58.66″E | 89 m | + |
| Narlıkuyu1 | 36°26′31.94″N, 34°5′57.47″E | 91m | + |
| Narlıkuyu1 | 36°26′34.42″N, 34°5′57.89″E | 98 m | + |
| Narlıkuyu1 | 36°26′39.23″N, 34°5′58.53″E | 108 m | + |
| Narlıkuyu1 | 36°26′38.67″N, 34°6′6.56″E | 97 m | + |
| Narlıkuyu1 | 36°26′38.27″N, 34°6′9.00″E | 88 m | + |
| Narlıkuyu1 | 36°26′27.61″N, 34°5′58.58″E | 89 m | + |

**Table 1.** (Continued).

| | | | |
|---|---|---|---|
| 18. Narlıkuyu, along the way | 36°25′48.93″N, 34°6′23.29″E | 42 m | - |
| 19. Narlıkuyu2 | 36°27′7.89″N, 34°6′59.17″E | 81 m | - |
| 20. Cumhuriyet | 36°28′42.89″N, 34°5′47.92″E | 328 m | - |
| 21. Kızkalesi1 | 36°27′55.35″N, 34°8′6.04″E | 46 m | + |
| Kızkalesi1 | 36°27′55.29″N, 34°8′6.23″E | 45 m | + |
| Kızkalesi1 | 36°27′53.17″N, 34°8′8.04″E | 34 m | + |
| 22. Kızkalesi2 | 36°28′6.71″N, 34°8′43.68″E | 42 m | + |
| Kızkalesi2 | 36°28′10.57″N, 34°8′41.23″E | 51 m | + |
| 23. Hüseyinler2 | 36°30′36.95″N, 34°5′54.24″E | 455 m | + |
| Hüseyinler2 | 36°30′38.01″N, 34°5′52.59″E | 457 m | + |
| 24. Ayaş | 36°29′33.21″N, 34°9′44.65″E | 117 m | + |
| Ayaş | 36°29′33.12″N, 34°9′39.44″E | 133 m | + |
| Ayaş | 36°29′33.87″N, 34°9′37.60″E | 139 m | + |
| Ayaş | 36°29′34.94″N, 34°9′34.30″E | 196 m | + |
| 25. Yemişkuyu | 36°30′0.80″N, 34°10′29.34″E | 70 m | + |
| 26. Cambazlı | 36°33′46.27″N, 34°2′15.80″E | 860 m | - |
| 27. 4 km W of Karaahmetli | 36°32′30.75″N, 34°5′12.08″E | 653 m | - |
| 28. Karaahmetli1 | 36°31′38.82″N, 34°7′26.52″E | 483 m | + |
| Karaahmetli1 | 36°31′45.06″N, 34°7′26.18″E | 502 m | + |
| Karaahmetli1 | 36°31′45.95″N, 34°7′26.60″E | 505 m | + |
| Karaahmetli1 | 36°31′48.36″N, 34°7′25.03″E | 510 m | + |
| Karaahmetli1 | 36°31′48.42″N, 34°7′25.05″E | 512 m | + |
| Karaahmetli1 | 36°31′48.48″N, 34°7′25.11″E | 513 m | + |
| 29. Karaahmetli2 | 36°33′25.79″N, 34°8′6.33″E | 630 m | - |
| 30. 2 NW of Kumkuyu | 36°31′20.64″N, 34°11′24.39″E | 162 m | + |
| 31. 2 km N of Kumkuyu | 36°31′47.89″N, 34°12′16.26″E | 123 m | + |
| 2 km N of Kumkuyu | 36°31′48.5″N, 34°12′16.14″E | 124 m | + |
| 32. Kumkuyu | 36°32′41.73″N, 34°10′48.68″E | 347 m | + |
| Kumkuyu | 36°32′46.51″N, 34°10′45.59″E | 364 m | + |
| Kumkuyu | 36°32′51.17″N, 34°10′35.28″E | 387 m | + |
| 33. Azimli | 36°33′3.75″N, 34°12′9.01″E | 229 m | + |
| Azimli | 36°33′3.43″N, 34°12′8.18″E | 229 m | + |
| Azimli | 36°33′3.11″N, 34°12′7.46″E | 229 m | + |
| Azimli | 36°33′2.17″N, 34°12′5.46″E | 232 m | + |
| Azimli | 36°33′1.74″N, 34°12′4.52″E | 235 m | + |
| Azimli | 36°33′5.28″N, 34°12′4.28″E | 238 m | + |
| 34. Batısandal1 | 36°33′47.80″N, 34°10′42.79″E | 407 m | + |
| 35. Batısandal2 | 36°34′10.31″N, 34°11′58.17″E | 320 m | + |
| Batısandal2 | 36°34′10.87″N, 34°11′58.40″E | 321 m | + |
| 36. 4 km N of Batısandal | 36°35′21.37″N, 34°9′51.93″E | 591 m | - |
| 37. Merkez village | 36°34′31.27″N, 34°13′59.28″E | 153 m | - |
| 38. Çiriş | 36°36′35.07″N, 34°12′24.05″E | 481 m | - |
| 39. Gücüş | 36°38′8.91″N, 34°9′28.75″E | 802 m | - |

2.5 km², whereas that of the continuous population (population B) is about 102 km². However, more finely tuned estimation studies are necessary to determine the distribution area more precisely. Such a finely tuned sampling may also produce the extent of occurrence (EOO) value. On the other hand, there is no doubt that gaps in distribution are the result of urbanization, which reduces the total area of the species distribution and may cause further fragmentation.

All previous records for this species were from the area between 5 km east and 25 km north east of Silifke, and from localities close to the sea, along the motorway between Silifke and Mersin (Lehmann, 1966, 1969; Morlok, 1978; Spitzenberger, 1978; Macholán et al., 1995; Kıvanç et al., 1997, 2013; Musser and Carleton, 2005; Arslan et al., 2008; Kryštufek and Vohralík, 2009; Frynta et al., 2010). We recorded samples from sea level up to 510 m a.s.l. IUCN assessment has a distribution map with 17.6 km² for *Acomys cilicicus* (Amori et al., 2008). However, when we excluded the areas shown inside the Mediterranean, the net area of distribution in the IUCN map was calculated as 15.4 km² and our study extended the area to 104.5 km². On the other hand, we found that the distribution of the species does not cross westward of the Göksu River and that unsuitable habitat type and urbanization also limit the distribution area in the east around Limonlu.

### 3.2. Habitat

Most of the samples were collected from a mosaic of rocky habitat and maquis vegetation. A few specimens were trapped from stony walls between agricultural fields. This finding shows that the Turkish spiny mouse also inhabits rocky walls among fields, which is a new habitat recorded for the species, as well as the previously recorded Mediterranean brushy vegetation with rocky ground.

### 3.3. Cooccurring rodent species

We set a total of 3243 traps during the study and collected 41 *Acomys cilicicus* and 69 *Apodemus mystacinus* samples. According to our sampling results, in our sampling localities the Turkish spiny mouse shared the habitat only with *Apodemus mystacinus*. In contrast, Kryštufek and Vohralík (2009) indicated that *Mus domesticus* and *Mus macedonicus* were also present in the same area.

### 3.4. Population density

There are very limited data about the population density of *Acomys cilicicus* in Turkey. Kryštufek and Vohralík (2009) collected 22 small mammals in 1993, 16 (73%) of which were Turkish spiny mice, ten of them captured in a small area in a single night. Later Kıvanç et al. (1997) collected as many as 30 (83%) Turkish spiny mice and six *Apodemus mystacinus* (17%) from a single locality 20 km east of Silifke in 3 days using a total of 140 traps. The TNI was 21.43 for *Acomys cilicicus*. During the study 39 localities between 10 km west of Taşucu and 6 km west of Erdemli were studied using 3243 traps. The total number of Turkish spiny mice (*Acomys cilicicus*) collected during our study in 14 localities was 41 (37%) and the total number of broad-toothed field mice (*Apodemus mystacinus*) in 18 localities was 69 (63%). Using these data, the TNI was 1.10 for *Acomys cilicicus* and 1.85 for *Apodemus mystacinus*. If localities without trapping were excluded, the TNI was 3.41 for *Apodemus mystacinus* and 2.75 for *Acomys cilicicus* (Table 2). We also found that the abundance of *Acomys cilicicus* is higher in coastal areas and lower in inland sites (Figure 2).

It can be said that the population density of the Turkish spiny mouse has reduced from about 73%–83% to about 37% during the last 20 years by comparison with previous collection data. In the same period the TNI for *A. cilicicus* has reduced from about 21.42 (Kıvanç et al., 1997) to 2.75 (this study). However, it is necessary to keep in mind that this argument depends on few studies, and further studies may supply better results. The density and relative abundance depend greatly on additional exogenous and endogenous factors. There is no doubt that *A. cilicicus* also has frequent and short-term fluctuations that can depend on the amount of precipitation, the periods with the lowest and highest temperatures, the periods of aridity, the density of predators, etc. To make a conclusion on relative abundance, there is a need for prolonged studies (monitoring) of at least 3 years in order to identify and correlate these factors.

### 3.5. Main threats and IUCN status

The IUCN has listed threat factors within 12 categories (IUCN, 2017). The threats that affect Turkish spiny mice can be placed in seven of them as follows: Residential & commercial development, Agriculture & aquaculture,

**Table 2.** Trapping results and trap night index (TNI) for Sherman traps results.

| Species | Locality number surveyed | Locality number where specimens of the species collected | Trap number set | Total sample size collected | TNI = number of record/ 100 trap nights |
|---|---|---|---|---|---|
| *Acomys cilicicus* | 39 | 14 | 1490 | 41 | 2.75 |
| *Apodemus mystacinus* | 39 | 19 | 2025 | 69 | 3.41 |

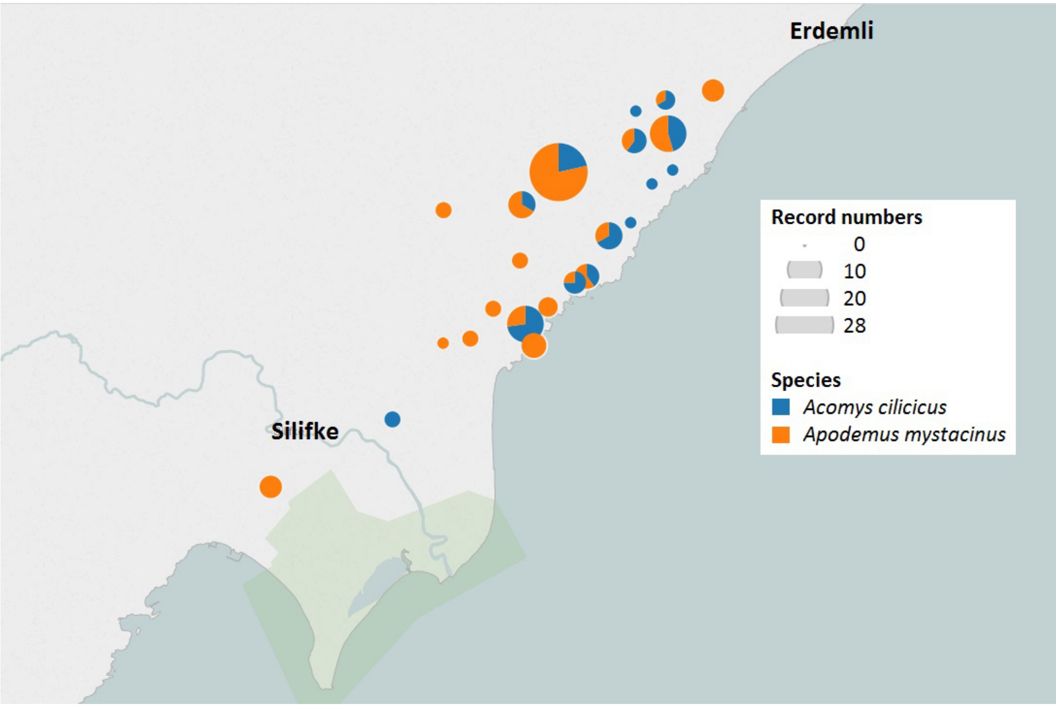


**Figure 2.** The distribution and relative abundance of *Acomys cilicicus* and *Apodemus mystacinus* samples in study area.

Energy production & mining, Transport & service corridors, Human intrusions & disturbance, Natural system modifications, and Pollution.

Residential & commercial development: In this category, increasing housing and urban areas in the spiny mouse's habitat are affecting the population now and this effect will continue in the future. The spiny mouse habitat extends along the Mediterranean coastal line and housing along the coastal line is continuously increasing. Such housing activities occupy the habitat and cause habitat loss. Tourism activities and recreation areas also continue to expand in the area and cause habitat loss. An area of 15.1 ha inside the IUCN distribution map of *A. cilicicus* seems to have been occupied by housing activities during the last 13 years (i.e. no. 6 in Figures 3 and 4).

Agriculture & aquaculture: Farmers are converting maquis areas to agricultural areas and such habitat change and agricultural activities cause habitat loss for Turkish spiny mice (i.e. nos. 1, 2, and 5 in Figures 3–5).

Energy production & mining: Some quarries in the area cause severe habitat destructions (see no. 4 area in Figures 3 and 4).

Transport & service corridors: The newly constructed motorway along the coastline destroyed the areas most populated by Turkish spiny mice between Susanoğlu and Akdeniz. The area of occupancy of the motorway here is calculated as 0.674 km$^2$ along 19 km. There is no doubt that in the effect area of the motorway, because of smog pollution, light pollution, and noise pollution, Turkish spiny mice could have disappeared from a larger area. Most historical records of *A. cilicicus* were from near the motorway that extends along the coastline. That is why, even if the total area is less than 1 km$^2$, it could have caused severe effects along the motorway. This part of the road is completely located inside the distribution area of *A. cilicicus* given in the IUCN distribution map. The length of the distribution area in the IUCN map is about 8.35 km and 6.23 km of the newly constructed motorway is located in this area. Besides the highway along the coastline, some motorways also are being constructed in natural *A. cilicicus* habitats and cause habitat loss (see no. 2 area in Figures 3 and 5).

Human intrusions & disturbance: The coastline along the motorway has dense human settlements and human pressure on the habitats around it. Satellite images from 13 years ago and today clearly show the increase in human intrusions on Turkish spiny mouse habitats (Figures 4 and 5).

Natural system modifications: Motorway constructions, housing activities, conversion of maquis areas to agricultural fields, and quarries in natural habitats are all modifications that affect *A. cilicicus* distribution in the area (Figures 4 and 5).

Pollution: Motorway constructions and increasing traffic density, increasing settlements, increasing agricultural activities with the opening of new agricultural

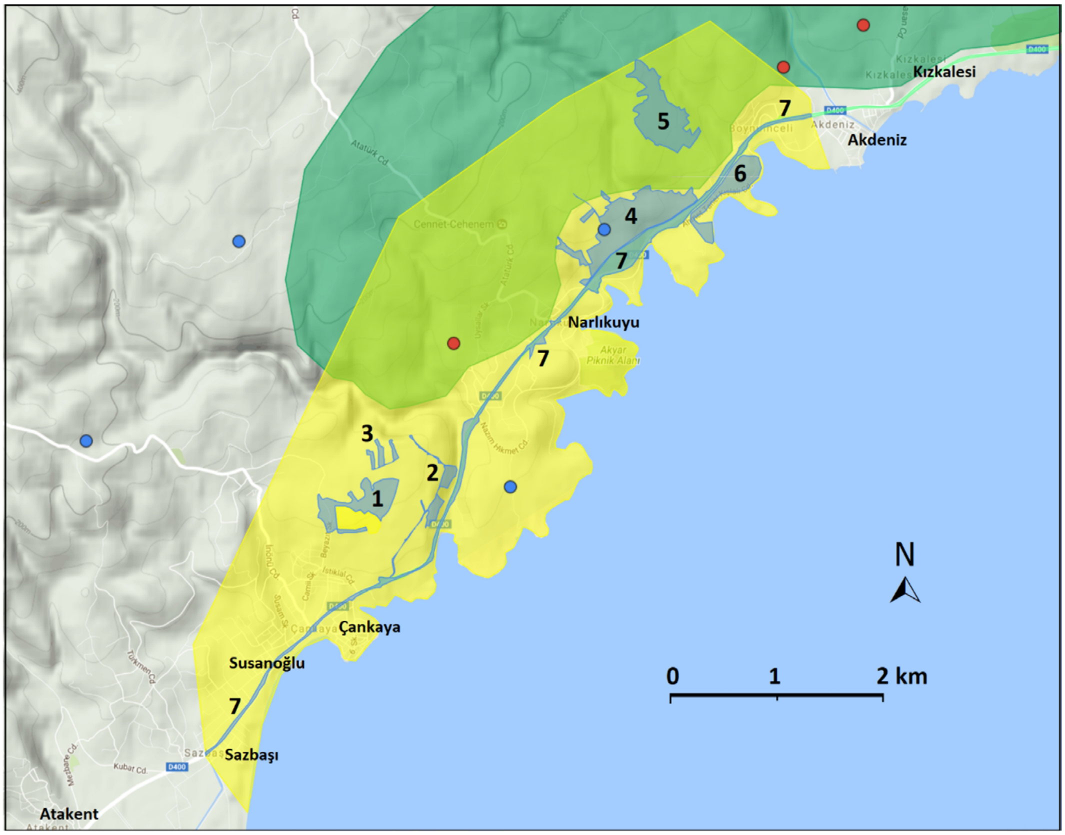


**Figure 3.** IUCN distribution map of *Acomys cilicicus* (yellow dashed area). Blue dashed areas (1 to 7 on map) show destroyed/changed habitats between 2004 and 2017.

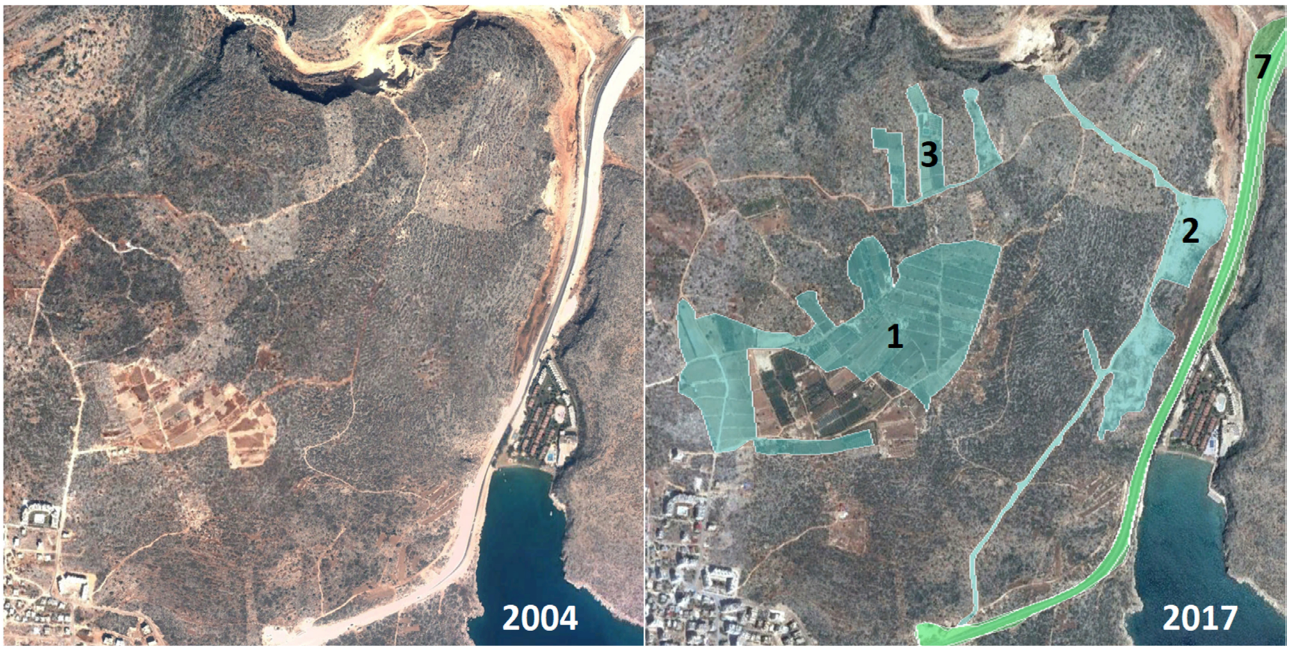


**Figure 4.** Habitat change in the distribution area of *Acomys cilicicus*. Blue and green dashed areas (1, 2, 3, and 7 on map) show destroyed/changed habitats between 2004 and 2017.

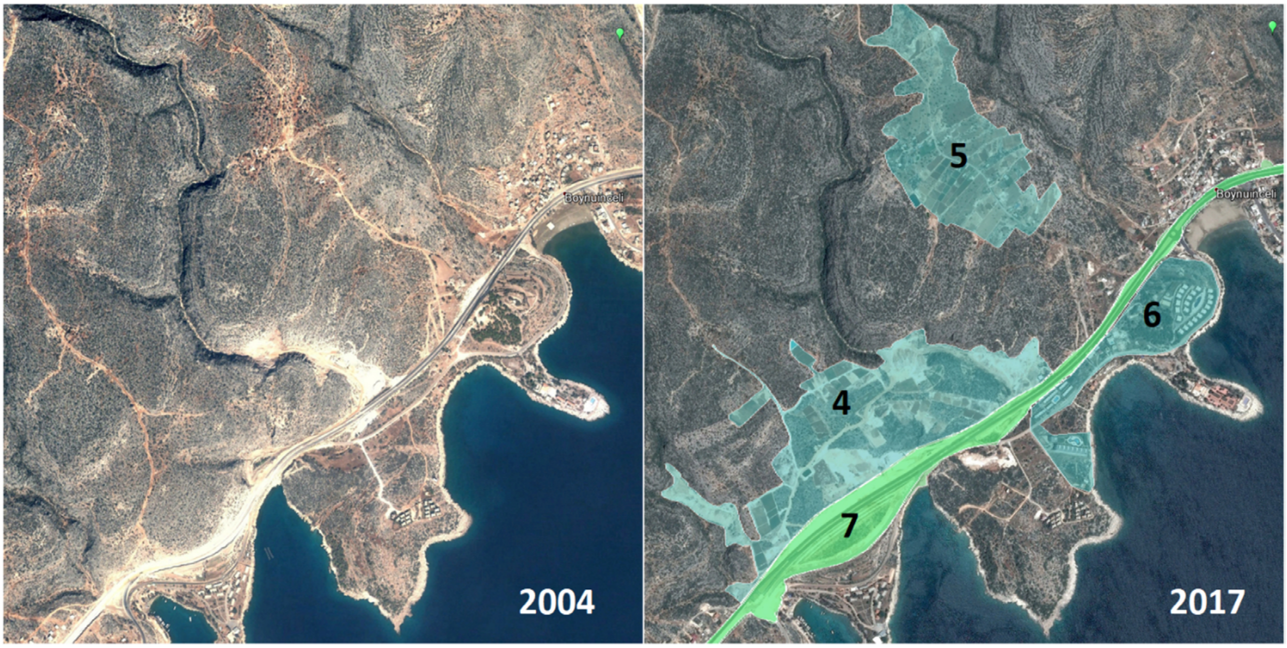


**Figure 5.** Habitat change in the distribution area of *Acomys cilicicus*. Blue and green dashed areas (4–7 on map) show destroyed/changed habitats between 2004 and 2017.

areas, and quarries in natural areas inevitably increase pollution in the *A. cilicicus* habitats surrounding these areas.

As we have presented, the main threats for this species are urbanization, habitat loss because of highway construction, conversion of Mediterranean brushy areas to agricultural fields, forestry applications including the planting of pine seedlings in maquis vegetation, habitat fragmentations, isolated small populations, quarries, and new motorways through spiny mouse habitats. These threats are especially harsh in the coastal areas where *Acomys cilicicus* is distributed. As the density of the species is higher in these coastal areas, negative effects may cause severe damage for the population in the near future. The occupied parts (nos. 1–7 in Figure 3) inside the distribution area given by the IUCN (Figure 3) measure about 152 ha. The total distribution area given by the IUCN is about 15.4 $km^2$ and habitat loss constitutes about 10% of this area. There is no doubt that impact areas of the occupied parts are much larger.

According to Kryštufek and Vohralík (2009), during their visit to the habitat of the Cilician spiny mouse in 2004, they found the habitat partly destroyed due to road reconstruction work and stated that the entire area is under rapid expansion of urbanization.

The IUCN evaluation of *Acomys cilicicus* in 2006 listed the species as CR because of the very small distribution range and threat factors that affect the species. However, it was reevaluated in 2008 and was listed as DD because of taxonomic issues (IUCN, 2017). We reevaluate the species here by using data collected by us and given in literature after 2008. According to the IUCN's criteria to evaluate the IUCN status of a species, Criterion A seems to be suitable for *Acomys cilicicus*. TNI results show that there has been an 87% reduction in the *Acomys* population size during the past 20 years (A2b). This is a rough estimation because of insufficient data in the past. However, the threats and habitat loss mentioned here imply that population reduction is undoubtedly taking place in the distribution area. The main threats that cause this reduction are urbanization, conversion of maquis areas to agricultural fields, quarries, motorways through the maquis areas, and the newly established double-lane motorway along the coastal zone. As these threats have become more severe in the last 10 years, it is not difficult to assume that most of the population loss could have happened in the last decade. According to status A2b, the species should be listed as CR. If the distribution area of occupancy of the species is less than 5000 $km^2$ (B1), and additionally, if it falls into at least two of three conditions listed as a, b, and c, then the species could be listed as CR. In the case of *Acomys cilicicus*, first the EOO of the species is less than 5000 $km^2$ (about 104 $km^2$), and continuing decline is observed in the area of occupancy, as well as in the number of mature individuals (b), because of the newly established double-lane motorway along the coastal zone, urbanization, deforestation, farming areas opening among rocky habitats, quarries, etc. Trapping success is an estimate of population, and the TNI is a measure of density. These values indicate a serious decline in the past 20 years. It is likely that most of this decline happened in the last 10 years (A2b and A2bc) because of urbanization, double-lane motorway construction, farming areas opening among rocky habitats, and quarries in the area. The species also deserves the CR status that meets criterion B1b because of the continuing decline observed as a result of the threats that affect the population.

The relative abundance of Turkish spiny mice among other sympatric small mammals has reduced from about 73% (1995 data in Kryštufek and Vohralík, 2009) to 83% (Kıvanç et al., 1997) to about 37% (this study) in the last 20 years. In the same period TNI has reduced from about 21.42 (Kıvanç et al. 1997) to 2.75 (this study). This change show about an 87% decline in density in the last 20 years. It should be borne in mind that the data prepared in one collection day given by Kryštufek and Vohralík (2009) and three collection days given by Kıvanç et al. (1997) could not directly represent the total density value in the distribution area of *A. cilicicus* at that time. However, these are the only density data available, and in the sampling period of our study the high density values recorded by Kıvanç et al. (1997) and Kryštufek and Vohralík (2009) were not observed at any sampling point or location. Thus, a serious decline in density is clear. The values show that a continuous and serious decline of the Turkish spiny mouse has occurred. The total distribution area and the decline in the population of Turkish spiny mice meets the critically endangered (CR) status of the IUCN, so the proposed IUCN status for *Acomys cilicicus* is CR. As explained here, the status of the species meets IUCN criteria A2b, A2c, and B1b.

### 3.6. Conservation

The distribution area of the species in Turkey is very restricted. Additionally, this restricted area is under serious anthropogenic pressure because of urbanization, habitat loss because of highway construction, conversion of Mediterranean brushy areas to agricultural fields and quarries, and new motorways. Current distribution data show that the species now has two isolated populations, and one of the populations is very small. It is likely that these small populations will disappear very quickly. The first step in conservation should be to connect these two subpopulations by protecting and recovering the habitats between them and allowing animals distributed in the area to connect to each other. Secondly, all suitable habitat in the general distribution area should be firmly protected and any kind of habitat alteration should be constrained. Dense populations close to the seashore should be determined in detail. These populations should be protected, and some way to connect them to main populations on the other side of the highway should be found. The typical habitat of the species is rocky habitat with maquis vegetation. No samples were captured from pine forest or pine plantation areas. Therefore, maquis scrublands in the area must be strictly protected and forestry applications should not be implemented. All small isolated populations should be determined and necessary precautions should be taken to connect them to the main population to prevent these small populations from suffering genetic drift. Additionally, for a better evaluation, a species conservation action plan should be prepared for this species as soon as possible.

### 3.7. Conclusions

The results presented here show that the Turkish spiny mouse, *Acomys cilicicus*, is one of the smallest-ranging endemic mammal species in Turkey. The distribution area of the species is now under severe human pressure, especially because of urbanization and its side effects. As a result of negative effects, the population density of the species seems to have declined in the last 20 years from 21.42 to 2.75 as measured by TNI. The small distribution range and steep decline in the population mean that the IUCN status of the species should be CR. Additionally, this decline in population density and fragmentation of the population into isolates means that serious precaution measures are needed to protect the species.

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